

Levi Felix, PhD

Postdoctoral Research Associate

Department of Materials Science and Nanoengineering

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Education

- 2018 – 2022  **Ph.D., State University of Campinas - UNICAMP** in Physics.
Thesis title: *Computational Mechanics of Porous and Two-dimensional Carbon Nanomaterials*.
- 2016 – 2018  **M.Sc., Federal University of Ceará - UFC** in Physics.
Dissertation title: *Wave-packet propagation in phosphorene in the presence of grain boundaries*.
- 2013 – 2016  **B.S., Federal University of Ceará - UFC** in Physics (Magna Cum Laude).
Monography title: *Landau levels in graphene influenced by one dimensional potentials*.

Skills

- Softwares  LAMMPS, SIESTA, Quantum ESPRESSO, CP2K, VASP, DFTB+, VMD, VESTA and OVITO.
- Coding  Python, C++ and L^AT_EX.
- Methods  Molecular Dynamics Simulations, Density Functional Theory, Density Functional Tight-Binding, Nudged-Elastic Band, Umbrella Sampling, Metadynamics.

Awards and Honors

- 2022  **Recognition Award from NIST** for oral presentation on *Symposium DS02: Advanced Manufactured Materials—Innovative Experiments, Computational Modeling and Applications* at MRS Spring Meeting 2022 in Honolulu-HI, United States.
- 2021  **Best Poster Award** on *Symposium EQ20: Beyond Graphene 2D Materials Synthesis, Properties and Device Applications* at MRS Fall Meeting 2021 in Boston-MA, United States.
- 2016  **Magna Cum Laude** on *Bachelor of Physics* at Federal University of Ceará in Fortaleza-CE, Brazil.

Conferences and Proceedings

- 2022  **MRS Spring Meeting.** Oral presentation. Honolulu-HI, United States.
- 2021  **MRS Fall Meeting.** Poster presentation. Boston-MA, United States.
- 2019  **MRS Fall Meeting.** Poster presentation. Boston-MA, United States.
 **MRS Brazil Meeting.** Poster presentation. Balneário Camboriú-SC, Brazil.

Employment History

- Oct 2022 – Present  **Postdoctoral Research Associate.** Rice University, Department of Materials Science and Nanoengineering
- Aug 2019 – Dec 2019  **Graduate Teaching Assistant.** State University of Campinas (Unicamp), 'Gleb Wataghin' Institute of Physics. General Physics IV
- Aug 2017 – Dec 2017  **Graduate Teaching Assistant.** Federal University of Ceará (UFC), Physics Department. Laboratory of Electricity and Magnetism I

Languages

Portuguese	📖	Native
English	📖	Fluent
Spanish	📖	Intermediate

Publications

- 1 Felix, L. C., Penev, E. S., Sun, M., & Yakobson, B. I. (2026). Topochemistry of ammonium perchlorate: How surface morphology mediates its sublimation. *The Journal of Physical Chemistry Letters*.
🔗 doi:<https://doi.org/10.1021/acs.jpclett.6c01896>
- 2 Felix, L. C., Pereira, M. L., Fabris, G. S., Fonseca, A. F., & Galvão, D. S. (2026). Kapitza resistance and thermal transport in knotted graphene nanoribbons. *Applied Surface Science*, 167187.
🔗 doi:<https://doi.org/10.1016/j.apsusc.2026.167187>
- 3 Felix, L. C., Li, Q.-K., Penev, E. S., & Yakobson, B. I. (2025). Ab Initio Molecular Dynamics Insights into Stress Corrosion Cracking and Dissolution of Metal Oxides. *Materials*, 18(3), 538.
🔗 doi:[10.3390/ma18030538](https://doi.org/10.3390/ma18030538)
- 4 Felix, L. C., Ambekar, R., Tromer, R. M., Woellner, C. F., Rodrigues, V., Ajayan, P. M., ... Galvao, D. S. (2024). Schwarzites and triply periodic minimal surfaces: From pure topology mathematics to macroscale applications. *Small*, 2400351. 🔗 doi:[10.1002/sml.202400351](https://doi.org/10.1002/sml.202400351)
- 5 Li, H., Felix, L. C., Li, Q., Ruan, Q., Yakobson, B. I., & Hersam, M. C. (2024). Atomic-resolution vibrational mapping of bilayer borophene. *Nano Letters*, 24(34), 10674–10680.
🔗 doi:[10.1021/acs.nanolett.4c03224](https://doi.org/10.1021/acs.nanolett.4c03224)
- 6 Tromer, R. M., Felix, I. M., Felix, L. C., Machado, L. D., Woellner, C. F., & Galvao, D. S. (2024). Hydrogen atom/molecule adsorption on 2d metallic porphyrin: A first-principles study. *Chemical Physics*, 577, 112142. 🔗 doi:<https://doi.org/10.1016/j.chemphys.2023.112142>
- 7 Tromer, R. M., Felix, L. C., Baughmann, R. H., Galvao, D. S., & Woellner, C. F. (2024). Transforming two-dimensional carbon allotropes into three-dimensional ones through topological mapping: The case of biphenylene carbon (graphenylene). *The Journal of Physical Chemistry A*, 128(35), 7346–7352.
🔗 doi:[10.1021/acs.jpca.4c01339](https://doi.org/10.1021/acs.jpca.4c01339)
- 8 de Alcântara, A. C. S., Felix, L. C., Galvão, D. S., Sollero, P., & Skaf, M. S. (2023). The role of the extrafibrillar volume on the mechanical properties of molecular models of mineralized bone microfibrils. *ACS biomaterials science & engineering*, 9(1), 230–245.
🔗 doi:[10.1021/acsbiomaterials.2c00728](https://doi.org/10.1021/acsbiomaterials.2c00728)
- 9 Moreira, M., Felix, L. C., Cottancin, E., Pellarin, M., Ugarte, D., Hillenkamp, M., ... Rodrigues, V. (2023). Influence of cluster sources on the growth mechanisms and chemical composition of bimetallic nanoparticles. *The Journal of Physical Chemistry C*. 🔗 doi:[10.1021/acs.jpcc.2c08044](https://doi.org/10.1021/acs.jpcc.2c08044)
- 10 Felix, L., Ambekar, R. S., Woellner, C., Kushwaha, B., Pal, V., Tiwary, C. S., & Galvao, D. S. (2022). Mechanical properties of 3D-printed pentadiamond. *Journal of physics D: Applied physics*.
🔗 doi:[10.1088/1361-6463/ac91dc](https://doi.org/10.1088/1361-6463/ac91dc)
- 11 Felix, L. C., & Galvao, D. S. (2022a). Guided fractures in graphene mechanical diode-like structures. *Phys. Chem. Chem. Phys.* 🔗 doi:[10.1039/D2CP01207C](https://doi.org/10.1039/D2CP01207C)
- 12 Alcântara, A. C. S., Felix, L. C., Galvão, D. S., Sollero, P., & Skaf, M. S. (2022). Devising bone molecular models at the nanoscale: From usual mineralized collagen fibrils to the first bone fibers including hydroxyapatite in the extra-fibrillar volume. *Materials*, 15(6). 🔗 doi:[10.3390/ma15062274](https://doi.org/10.3390/ma15062274)

- 13 Felix, L. C., Tromer, R. M., Woellner, C. F., Tiwary, C. S., & Galvao, D. S. (2022a). Mechanical response of pentadiamond: A DFT and molecular dynamics study. *Physica B: Condensed Matter*, 629, 413576. [doi:10.1016/j.physb.2021.413576](https://doi.org/10.1016/j.physb.2021.413576)
- 14 Gaal, V., Felix, L. C., Woellner, C. F., Galvao, D. S., Tiwary, C. S., d'Ávila, M. A., & Rodrigues, V. (2021). Mechanical properties of 3D printed macroscopic models of schwarzites. *Nano Select*. [doi:10.1002/nano.202100147](https://doi.org/10.1002/nano.202100147)
- 15 Tromer, R. M., Felix, L. C., Ribeiro, L. A., & Galvao, D. S. (2021). Optoelectronic properties of amorphous carbon-based nanotube and nanoscroll. *Physica E: Low-dimensional Systems and Nanostructures*, 130, 114683. [doi:10.1016/j.physe.2021.114683](https://doi.org/10.1016/j.physe.2021.114683)
- 16 Tromer, R. M., Felix, L. C., Woellner, C. F., & Galvao, D. S. (2020a). A DFT investigation of the electronic, optical, and thermoelectric properties of pentadiamond. *Chemical physics letters*, 138210. [doi:10.1016/j.cplett.2020.138210](https://doi.org/10.1016/j.cplett.2020.138210)
- 17 Cunha, S. M., da Costa, D. R., Felix, L. C., Chaves, A., & Pereira, J. M. (2020). Electronic and transport properties of anisotropic semiconductor quantum wires. *Phys. Rev. B*, 102, 045427. [doi:10.1103/PhysRevB.102.045427](https://doi.org/10.1103/PhysRevB.102.045427)
- 18 Felix, L. C., Gaál, V., Woellner, C. F., Rodrigues, V., & Galvao, D. S. (2020). Mechanical Properties of Diamond Schwarzites: From Atomistic Models to 3D-Printed Structures. *MRS Advances*, 5(33), 1775–1781. [doi:10.1557/adv.2020.175](https://doi.org/10.1557/adv.2020.175)
- 19 Felix, L. C., Tromer, R. M., Autreto, P. A. S., Ribeiro Junior, L. A., & Galvao, D. S. (2020). On the mechanical properties and thermal stability of a recently synthesized monolayer amorphous carbon. *The Journal of Physical Chemistry C*, 124(27), 14855–14860. [doi:10.1021/acs.jpcc.0c02999](https://doi.org/10.1021/acs.jpcc.0c02999)
- 20 Tromer, R. M., Felix, L. C., Woellner, C. F., & Galvao, D. S. (2020b). On the structural stability and optical properties of germanium-based schwarzites: A density functional theory investigation. *Physical Chemistry Chemical Physics*, 22(28), 16286–16293. [doi:10.1039/d0cp02143a](https://doi.org/10.1039/d0cp02143a)
- 21 Felix, L. C., Woellner, C. F., & Galvao, D. S. (2019a). Mechanical and energy-absorption properties of schwarzites. *Carbon*. [doi:10.1016/j.carbon.2019.10.066](https://doi.org/10.1016/j.carbon.2019.10.066)